

ANN Session – 2

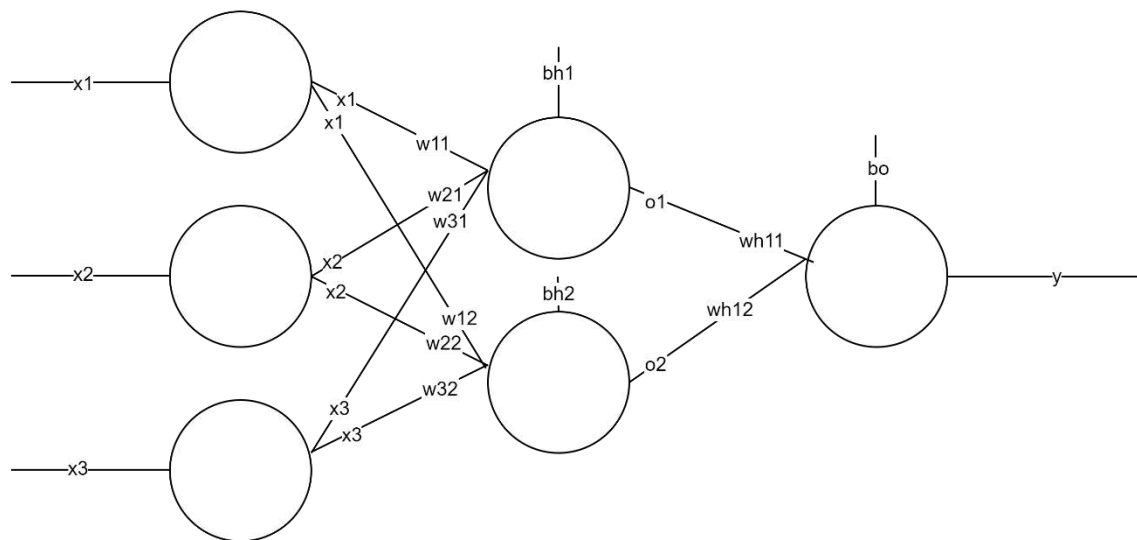
Example – 2:

3NN:

Input layer: 3 neurons

Hidden layer: 2 neurons

Output layer: 1 neurons



$$o1 = \text{Activation}(bh1 + w11 * x1 + w21 * x2 + w31 * x3)$$

$$o2 = \text{Activation}(bh2 + w12 * x1 + w22 * x2 + w32 * x3)$$

$$y = \text{Activation}(bo + wh11 * o1 + wh12 * o2)$$

Input layer outputs are = Inputs only

Hidden layer inputs are x_1, x_2, x_3 also $w_{11}, w_{12}, w_{21}, w_{22}, w_{31}, w_{32}$

$$o_1 = \text{Activation}(b_{h1} + w_{11} * x_1 + w_{21} * x_2 + w_{31} * x_3)$$

$$o_2 = \text{Activation}(b_{h2} + w_{12} * x_1 + w_{22} * x_2 + w_{32} * x_3)$$

How many weights at hidden layer: 6

number of neurons in input layer * number of neurons in hidden layer

How many parameters at hidden layer = weights + bias = 6 + 2 = 8

number of neurons in input layer * number of neurons in hidden layer
+ number of neurons in hidden layer

Matrix representation

$$o_1 = \text{Activation}(b_{h1} + w_{11} * x_1 + w_{21} * x_2 + w_{31} * x_3)$$

$$o_2 = \text{Activation}(b_{h1} + w_{12} * x_1 + w_{22} * x_2 + w_{32} * x_3)$$

$$\begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} b_{h1} \\ b_{h2} \end{bmatrix} = \begin{bmatrix} o_1 \\ o_2 \end{bmatrix}$$

Output layer

$$y = \text{Activation}(b_0 + w_{h11} * o_1 + w_{h12} * o_2)$$

How many weights at hidden layer: 2

number of neurons in hidden layer * number of neurons in output layer = $2 * 1 = 2$

How many parameters at hidden layer = weights + bias = $2 + 1 = 3$

number of neurons in hidden layer layer * number of neurons in output layer
+ number of neurons in output layer

Total parameters in above network = $8 + 3 = 11$

Maths Calculations

at Input layer:

- No bias – No weights : Total parameters = 0

At Hidden Layer:

Hidden layer – 1 : Neuron(1): $b_{h1} + w_{11} * x_1 + w_{21} * x_2 + w_{31} * x_3$

Hidden layer – 1 : Neuron(2): $b_{h2} + w_{12} * x_1 + w_{22} * x_2 + w_{32} * x_3$

Bias(2): b_{h1}, b_{h2} , Weights(6): $w_{11}, w_{21}, w_{12}, w_{22}, w_{31}, w_{32}$

Total parameters = 2 + 6 = 8

Trick: Hidden layer parameters

$inputlayer_{neurons} * Hiddenlayer_{Neurons} + Hiddenlayer_{Neurons}$

$inputlayer_{neurons} * Hiddenlayer_{Neurons}$: Number of weights

$Hiddenlayer_{Neurons}$: Number of Bias

At Output layer:

$o_1 = Activation(w_{h1} + w_{11} * x_1 + w_{21} * x_2)$

$o_2 = Activation(w_{h2} + w_{12} * x_1 + w_{22} * x_2)$

$y = Activation(w_o + w_{h1o} * o_1 + w_{h2o} * o_2)$

Trick: output layer parameters

$Hiddenlayer_{Neurons} * output_{layer} + output_{layer}$

$2 * 1 + 1 = 3$

4NN

Input layer = 3 neurons

Hidden layer – 1: 3 neurons

Hidden layer – 2: 2 neurons

output layer: 1 neuron

layer(neurons)	Parameter(weights + bias)
Input layer (3)	0
Hidden layer(3)	$3 * 3 + 3 = 12$
Hidden layer(2)	$3 * 2 + 2 = 8$
output layer(1)	$2 * 1 + 1 = 3$
	23

5NN

Input layer = 10 neurons

Hidden layer – 1: 10 neurons

Hidden layer – 2: 5 neurons

Hidden layer – 2: 3 neurons

output layer: 10 neuron

Total : 183 185 216, 222, 223, 282 , 183

<i>layer(nuerons)</i>	<i>Parameter(weights + bias)</i>
<i>Input layer (10)</i>	0
<i>Hidden layer(10)</i>	$10 * 10 + 10 = 110$
<i>Hidden layer(5)</i>	$10 * 5 + 5 = 55$
<i>Hidden layer(3)</i>	$5 * 3 + 3 = 18$
<i>output layer(10)</i>	$3 * 10 + 10 = 40$
	223

6NN

Input layer = 20 neurons

Hidden layer – 1: 15 neurons

Hidden layer – 2: 8 neurons

Hidden layer – 3: 5 neurons

Hidden layer – 4: 3 neuron

output layer: 10 neuron

Type equation here.

<i>layer(nuerons)</i>	<i>Parameter(weights + bias)</i>
<i>Input layer (20)</i>	0
<i>Hidden layer(15)</i>	$20 * 15 + 15 = 315$
<i>Hidden layer(8)</i>	$15 * 8 + 8 = 128$
<i>Hidden layer(5)</i>	$8 * 5 + 5 = 45$
<i>Hidden layer(3)</i>	$5 * 3 + 3 = 18$
<i>output layer(10)</i>	$3 * 10 + 10 = 40$
	546

Deep Neural Network

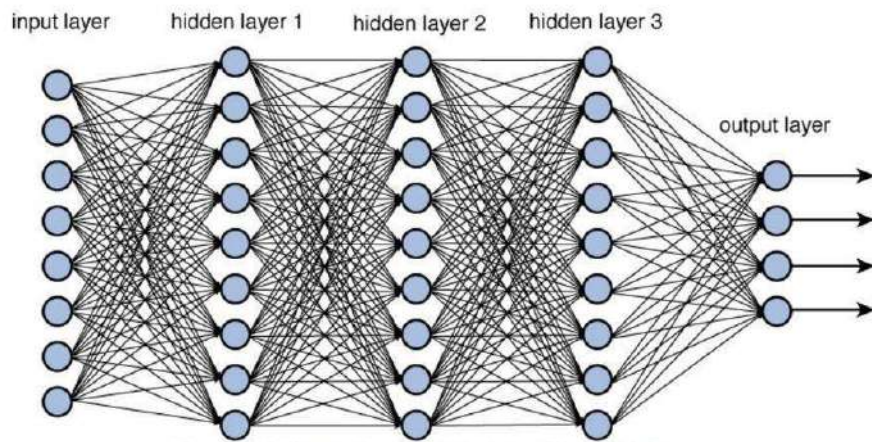


Figure 12.2 Deep network architecture with multiple layers.

- 1) *How to find the parameters*
- 2) *what is the activation function*
- 3) *How to choose the hidden layers*
- 4) *How to avoid the overfitting*
- 5) *How to speed up the process*